

NAME	ACTIVE	DRIVER	STATE	URL	SWARM	DOCKER	ERRORS
consul-node	-	virtualbox	Running	tcp://192.168.95.100:2376		v18.04.0-ce	
swarm-master	-	virtualbox	Running	tcp://192.168.95.101:2376	swarm-master (master)	v18.04.0-ce	
swarm-node1	-	virtualbox	Running	tcp://192.168.95.102:2376	swarm-master	v18.04.0-ce	

Figure 15: List of created virtual hosts

```
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
- 100% |*****| 612 0:00:00
```

Figure 16: Output from the Nginx server

```
Connecting to web2 (14.1.10.3:80)
<html><body><h1>It works!</h1></body></html>
- 100% |*****|
```

Figure 17: Output from the httpd server

1. Overlay Network 1: (overlay-net)

Go into one of the Swarm nodes using *ssh*:

```
$docker-machine ssh swarm-node1
$docker network create --driver overlay --subnet=13.1.10.0/24
overlay-net
```

2. Overlay Network 2: (overlay-net2)

Go into Swarm master using *ssh*:

```
$docker-machine ssh swarm-master
$docker network create --driver overlay --subnet=14.1.10.0/24
overlay-net2
```

Assign the subnet to the network of the Docker container. We have created two overlay networks. All the containers in the first overlay network named *overlay-net* are able to communicate with all other containers in that same overlay network. But, containers belonging to different overlay networks are not able to communicate with each other. We will also test if any container in the *overlay-net* network is able to connect to any other container in the same network, even if these are running on different hosts, but is not able to connect to a container in the *overlay-net2* network, even if these are running on the same host.

Testing multi-hosts and multi-networks

For testing the networks we have created, we need client and server terminology. So, here let us take an Nginx server with a simple Web page hosted on it and the client, which is the Docker container, inside or outside the network. If the client gets the hosted Web page on request, it shows that the client is in the same network of the Nginx server. If the server does not respond, it shows that client and Nginx server are in different

networks. To avoid installing and setting up the Nginx server, we use a Docker container for Nginx and directly run it. The following commands create a Nginx Docker container in Swarm master with the network *overlay-net* and run a Busybox container in Swarm-node1, also with the same network *overlay-net*.

```
$ docker-machine ssh swarm-master
$ docker run -itd --name=web --network=overlay-net
nginx:alpine
$ docker-machine ssh swarm-node1
$ docker run -it --rm --network=overlay-net busybox wget -O-
http://web
```

As both Nginx and Busybox containers are on the same network (*overlay-net*), Busybox gets the Nginx page. However, it will not be able to access the httpd page which is in the other network, even if both containers are running on the same host.

Let's create a httpd Docker container in Swarm-node1 with network *overlay-net2* and run a Busybox container in the Swarm master with the same network.

```
$ docker-machine ssh swarm-node1
$ docker run -itd --name=web --network=overlay-net2
httpd:alpine
$ docker-machine ssh swarm-master
$ docker run -it --rm --network=overlay-net2 busybox wget -O-
http://web
```

Figure 17 gives the output from the httpd container but not from the Nginx container.

This verifies our test of creating an overlay network on multiple hosts and accessing Docker containers using those networks. **END** 

References

- [1] Karl Matthias and Sean P Kane. Docker: Up and Running (2015). Introduction, pp. 1-5. Oreilly Media.
- [2] Outfitting an IT training environment with Docker containers at: <http://www.admin-magazine.com/Archive/2016/32/Outfitting-an-IT-training-environment-with-Docker-containers>
- [3] Empower Docker to Make Sufficient Infrastructure Available Effectively as well as Efficiently for Learning and Development, by Dr B. Thangaraju and Rajesh Srivastava, *Developer IQ Magazine*, March 2016, pp.54-60.
- [4] Multi-host networking with standalone swarms at <https://docs.docker.com/network/overlay-standalone.swarm/>
- [5] Use Overlay Networks at <https://docs.docker.com/network/overlay/>
- [6] Swarm Mode Overview at <https://docs.docker.com/engine/swarm/>
- [7] Docker Swarm Cluster using Consul at <http://blog.arungupta.me/docker-swarm-cluster-using-consul/>

By: Rajavaram Harika, Shreyak Upadhyay and Prof. B. Thangaraju

The authors are associated with the Open Source Technology Lab, International Institute of Information Technology, Bengaluru.